

Announcements 3/2

Exam 2 at 11 AM on Fri 3/20 week 7:

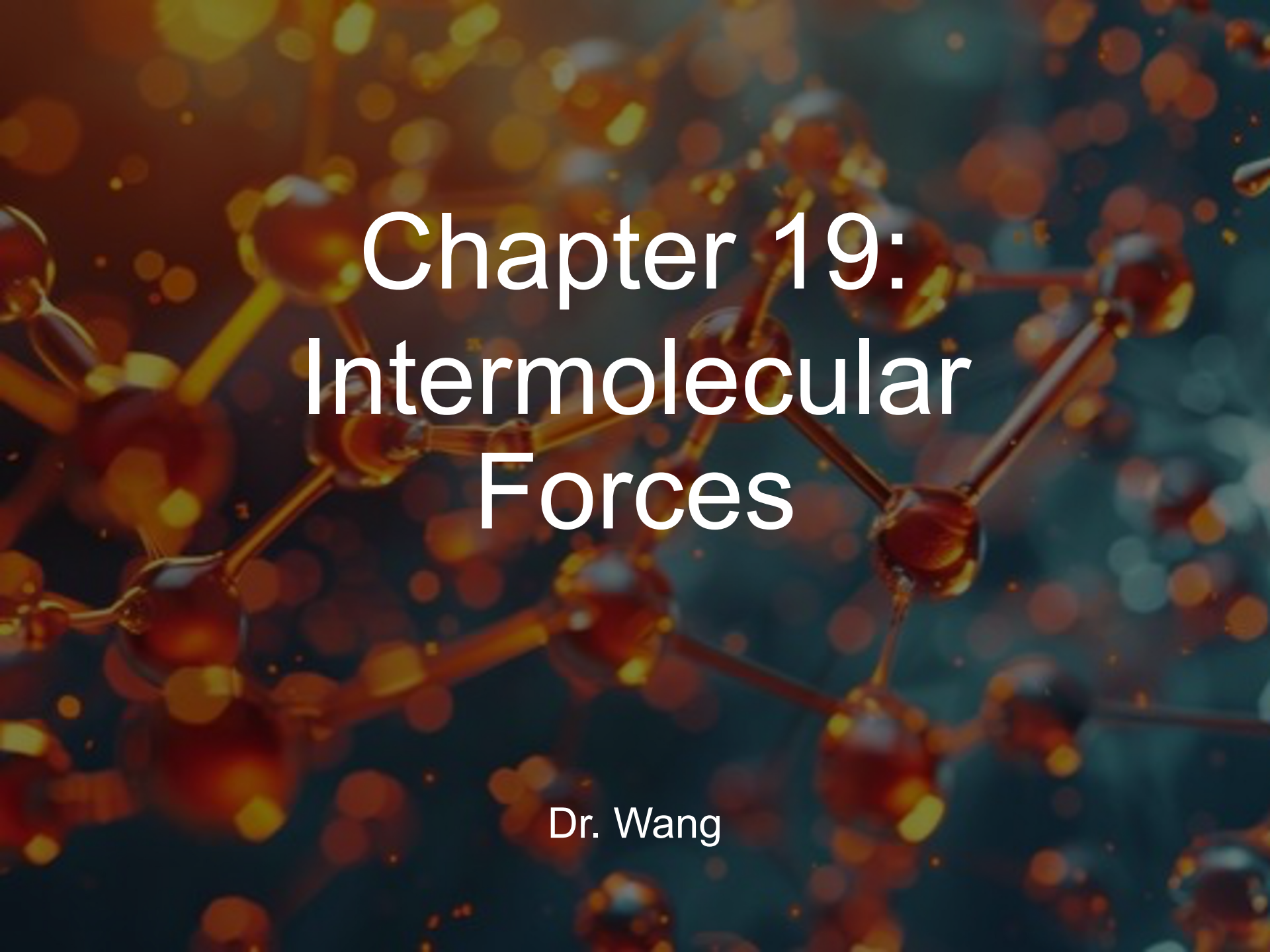
- Bonus point for Unit 2: 3 Chem RePS worksheets and Exam 1 Wrapper

Assignments that are due soon

- Pre-Class Assignment Week 5-W

Office Hour

- Dr. Wang's: Tue 10-11 AM at KC 230
- ~~Paige Kwan's: Mon 11:30 AM - 12:30 PM at the TLC (Demille 160P)~~
- Saige Douglass's: Tue ~~5:30-6:30~~ PM at the TLC (Demille 160P)
12:00 1:00 P/M

The background of the slide features a complex, three-dimensional molecular structure. It consists of numerous dark red, spherical atoms connected by thin, translucent orange-brown rods representing chemical bonds. The structure is dense and interconnected, filling the frame. The lighting is dramatic, with bright highlights on the atoms and bonds, creating a bokeh effect of out-of-focus light spots in shades of orange, yellow, and red against a dark, teal-blue background.

Chapter 19: Intermolecular Forces

Dr. Wang

Learning Objectives

- Draw Lewis structures geometrically for molecules with 1 central atom.
- Determine the molecular geometry of a molecule from the Lewis structure.
- Determine molecular polarity from the molecular geometry.

Lewis structures very often do not represent the real shape of molecules!

Compound:

Carbon dioxide

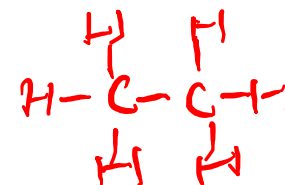
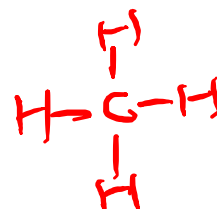
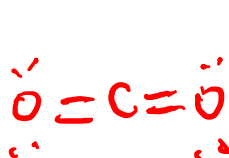
Methane

Molecular formula:

CO₂

CH₄

Lewis structure:

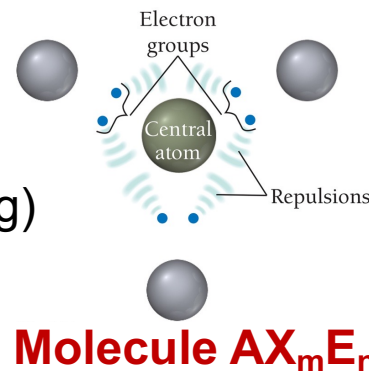


Valence Shell Electron Pair Repulsion Theory

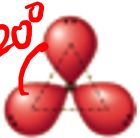
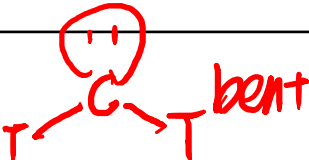
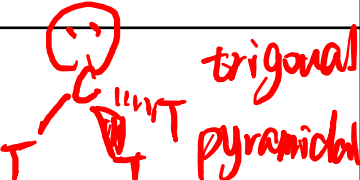
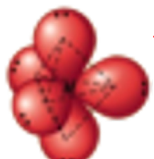
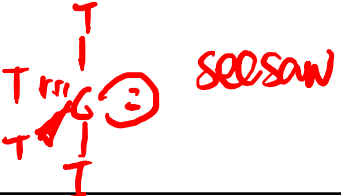
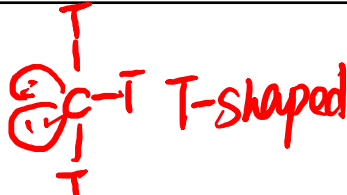
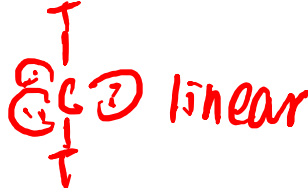
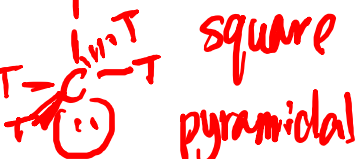
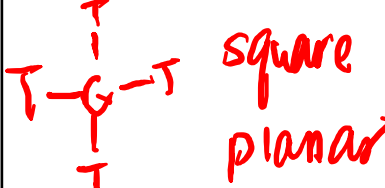
- Predict the geometry of central atom (A)
- Assumption: **electron groups** around central atom repels each other
 - Lone pairs (X) on central atoms : lp
 - Bonding pairs (E):

Atoms around the Central atom (**Independent to bonding**)

- Preferred geometry: minimization of repulsion

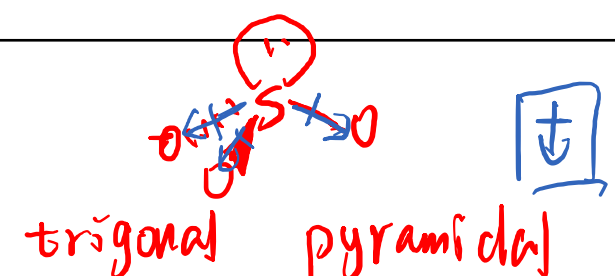
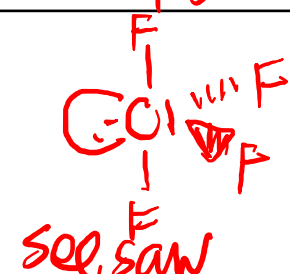


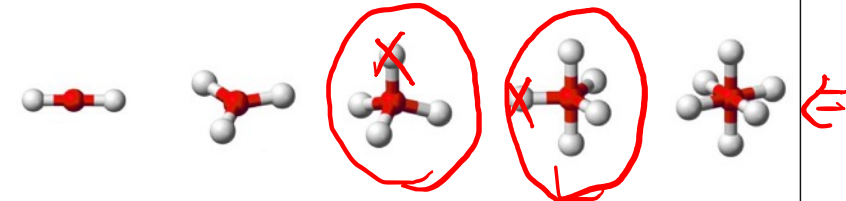
Molecular Geometries

# Electron Group	Molecular Geometry with 0 lone pair	Molecular Geometry with 1 lone pair	Molecular Geometry with 2 lone pairs	Molecular Geometry with 3 lone pairs
2	 180° linear			
3	 120° trigonal planar	 bent		
4 109.5°	 tetrahedral	 trigonal pyramidal	 bent	
5 120°	 trigonal bipyramidal	 seesaw	 T-shaped	 linear
6	 octahedral	 square pyramidal	 square planar	

Your Turn!

Determine the shape of SO_3^{2-} and XeF_4 .

	# Lone Pair	# Bonding Pair	# e-Groups	Molecular geometry in wedge-dash notation
$\left[\begin{array}{c} \text{:}\ddot{\text{O}}-\ddot{\text{S}}-\ddot{\text{O}}\text{:} \\ \\ \text{:}\ddot{\text{O}}\text{:} \end{array} \right]^{2-}$				 trigonal pyramidal
$\begin{array}{c} \text{F} \\ \\ \text{F}-\text{Cl}^+-\text{F} \\ \\ \text{F} \end{array}$				 see saw

	Possible Electron Geometry Figures	
	Possible Molecular Geometries	Bent, Linear, Octahedral, Seesaw, Square Planar, Square Pyramid, T-shaped, Tetrahedral, Trigonal Bipyramidal, Trigonal Planar, Trigonal Pyramidal

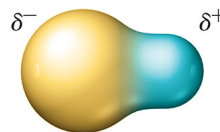
Bond Polarity

EN: H (2.20); C(2.55); O(3.44); Cl(3.16); Na (0.93)

- nonpolar Covalent Bond: $\Delta_{\text{EN}} \leq 0.4$
 - equally sharing of electrons

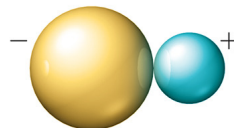


- polar Covalent Bond: $0.4 < \Delta_{\text{EN}} \leq 2.0$
 - unequally sharing of electrons
 - Larger the Δ_{EN} , the more polar the bond is.
 - Dipole moment: $\leftarrow \text{---} \text{---} \text{---} \rightarrow$



$\delta < 1$

- ionic Bond: $\Delta_{\text{EN}} > 2.0$
 - transferring of electrons

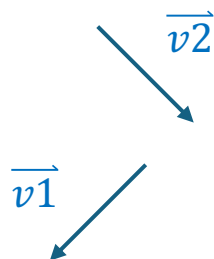


Molecular Polarity

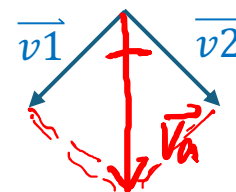
bond dipole + Molecular geometry

Dipoles cannot numerical added

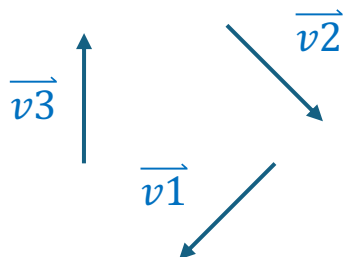
- Vector addition: add one by one using *parallelogram rule*
- Vectors with the same length but **opposite** direction **cancel out** each other



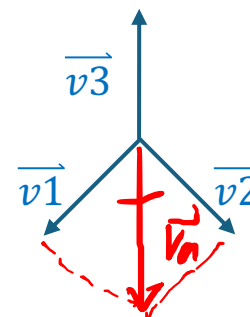
Move the head of the
vectors to the same point



$$\vec{v_1} + \vec{v_2} = \vec{v_a}$$



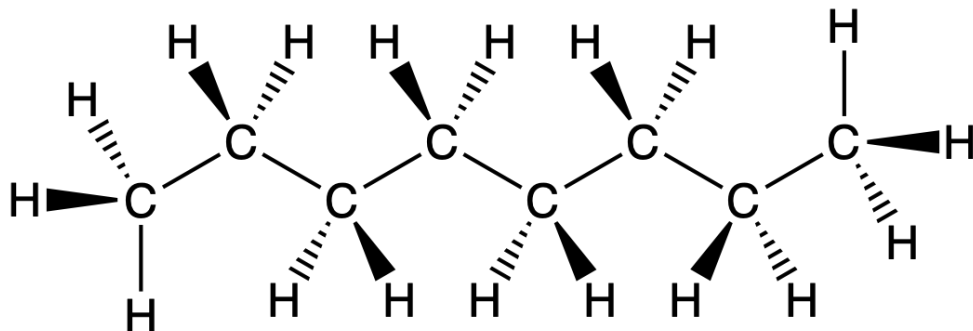
Move the head of the
vectors to the same point



$$\vec{v_1} + \vec{v_2} + \vec{v_3} = \vec{0}$$

Nonpolar Molecule

- Molecules with nonpolar bonds



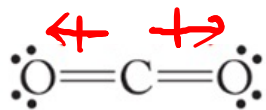
$$\Delta EN$$

$$C-C \quad \cancel{0} \quad \times$$

$$C-H \quad |2.55-2.20|$$

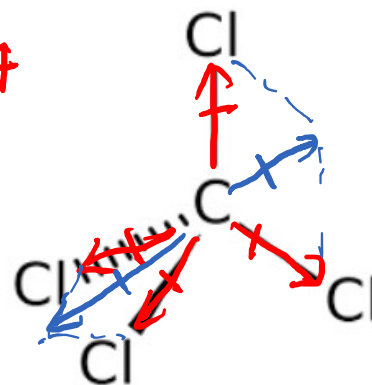
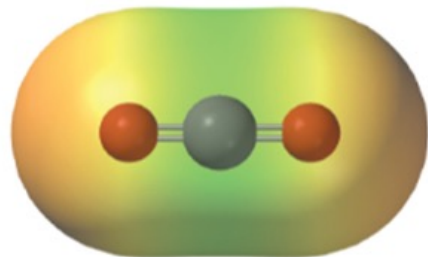
$$= 0.35 \quad \times$$

- Molecules with polar bonds but no net dipole
(Dipole moments can be cancelled out)



$$C=O \quad \Delta EN$$

$$|2.55-3.44| > 0.4$$



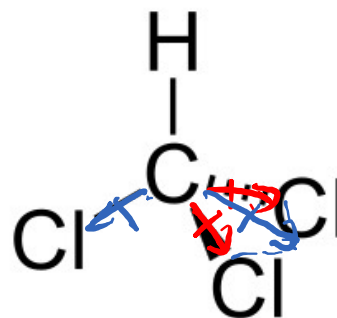
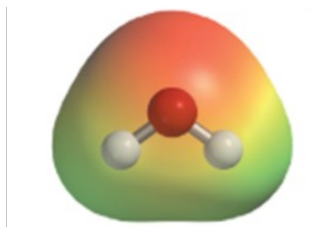
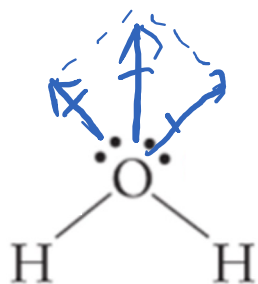
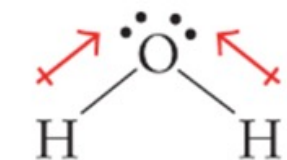
$$\Delta EN$$

$$C-Cl \quad |2.55-3.16|$$

$$> 0.4$$

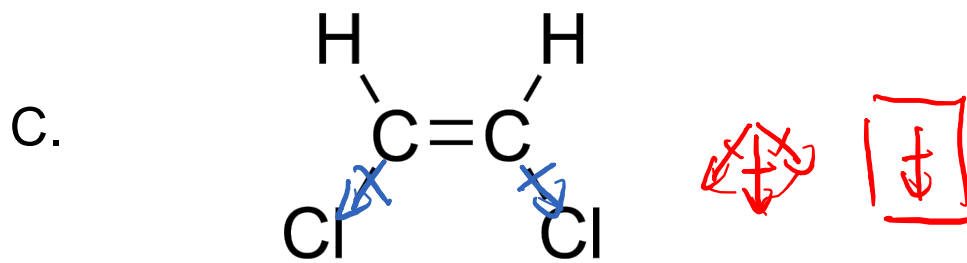
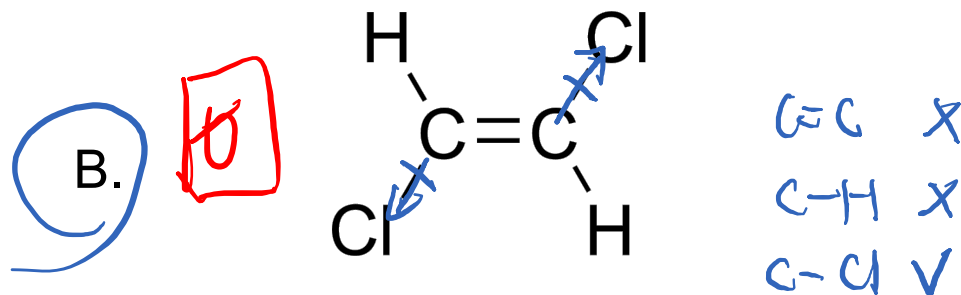
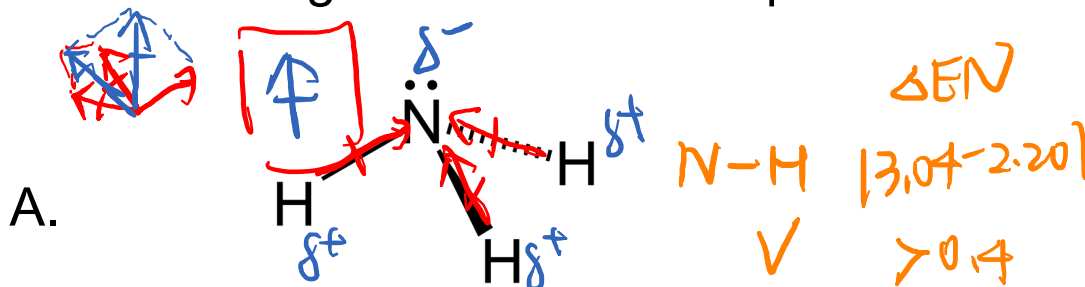
Polar Molecule

- Molecules with polar bonds and net dipole remains



Your Turn!

Which of the following molecule is a nonpolar molecule?

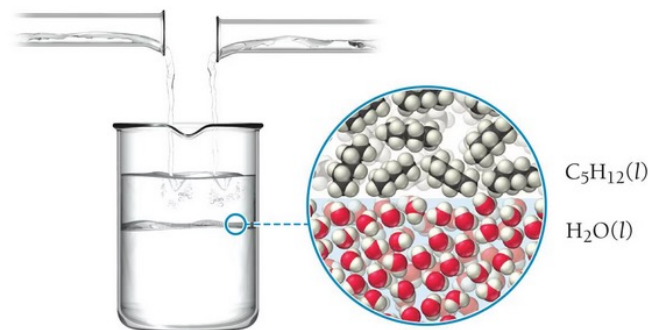


H				
2.20				
B	C	N	O	F
2.04	2.55	3.04	3.44	3.98
Al	Si	P	S	Cl
1.61	1.90	2.19	2.58	3.16
Ga	Ge	As	Se	Br
1.81	2.01	2.18	2.55	2.96
In	Sn	Sb	Te	I
1.78	1.96	2.05	2.1	2.66

Miscibility: Like Dissolves Like

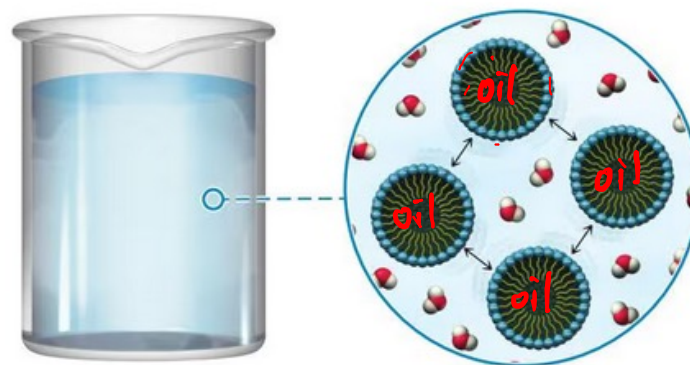
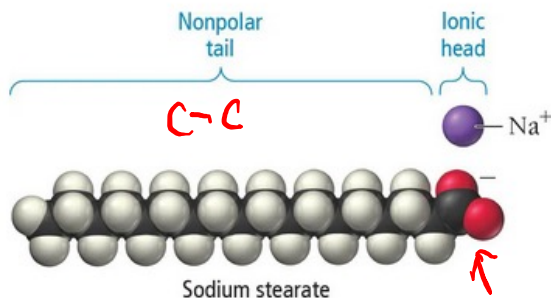


Polar and Nonpolar Substances



- **Polar** compounds dissolve polar compounds
- **Nonpolar** compounds dissolve nonpolar compounds
- **Polar** and **Nonpolar** do not dissolve in each other

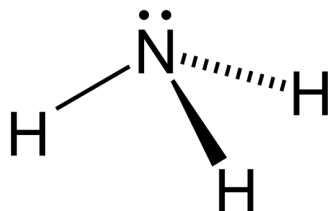
How does soap work?



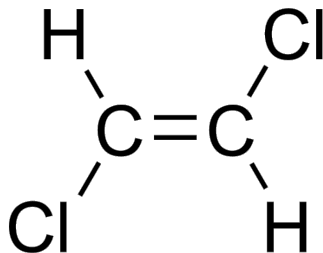
Your Turn!

Which of the following molecule(s) can be dissolved in water?

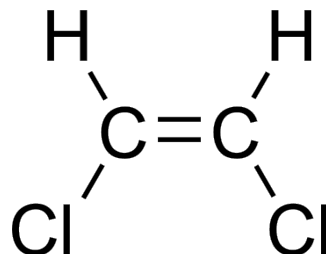
A.



B.



C.



H				
2.20				
B	C	N	O	F
2.04	2.55	3.04	3.44	3.98
Al	Si	P	S	Cl
1.61	1.90	2.19	2.58	3.16
Ga	Ge	As	Se	Br
1.81	2.01	2.18	2.55	2.96
In	Sn	Sb	Te	I
1.78	1.96	2.05	2.1	2.66

Exam 1 160/180 | bonus 1

Announcement 3/4

25 (V₂, V₃)

Exam 1 164/180 %

Bonus point for Unit 2: 3 Chem RePS worksheets and Exam 1 Wrapper

- 1st Chem RePS or SI worksheets for Unit 2 and Exam 1 Wrapper Assignment will be closed on 3/11 Wednesday 11:59 PM

Wed

Thu

Assignments that are due soon

- Pre-Class Assignment Week 5-F

Chem RePS Session

Wed 2:00 - 3:30 PM at KC 230

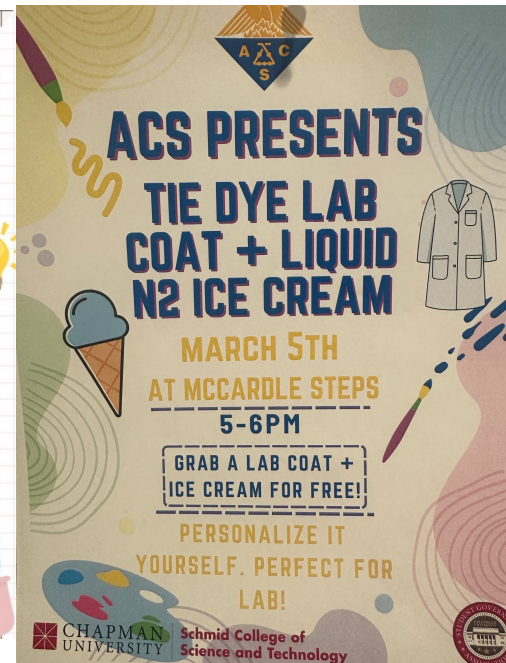
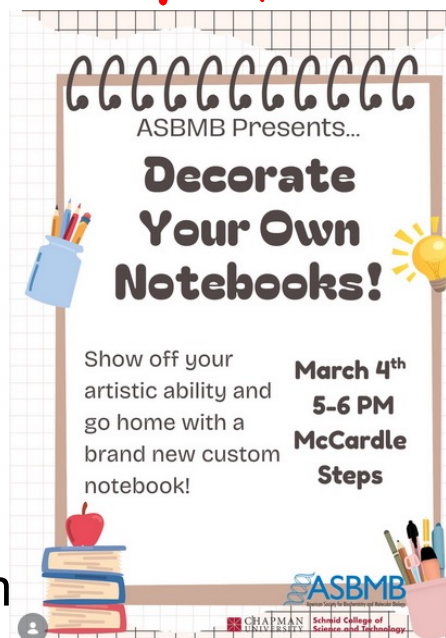
Office Hour

- Dr. Wang's: Thu 10-11 AM on Zoom

SI Session

Saige Douglass – Wed 5:30 – 7:30 PM at Hashinger 412

Paige Kwan - Thu 5 - 7 PM at Hashinger 412



Learning Objectives

- Explain why the IMF strengths are ranked in the following order: ion-dipole IMFs are the strongest, followed by H-bonding, followed by dipole-dipole attractions, and London Dispersion Forces have the weakest strength.
- Determine the Intermolecular forces (IMF) present for the molecules in a pure substance from the molecular structure
- Rank the relative strength of London Dispersion Forces for pure substances.
- Rank the relative strength of Dipole-Dipole attractions for pure substances.

Intermolecular vs Intramolecular Forces

- Intramolecular** forces (also known as chemical bond) hold atoms together in a molecule:

- ionic bonds
- covalent bonds

- Intermolecular** forces (IMF) are attractive forces between molecules

stronger ↓

<u>dispersion</u> forces <u>dipole-dipole</u> forces <u>hydrogen</u> bonding <u>ion-dipole</u> forces	} often referred to as van der Waals forces	All <u>electrostatic</u> by nature $E = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$
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- In general, **Intramolecular** forces is stronger than **Intermolecular** forces
- Breaking **Intramolecular** forces is a chemical change.
- Breaking **Intermolecular** forces is a physical change:

phase change; dissolution

(s) → (l) → (g) (s) → (l) → (g)

Intermolecular vs Intramolecular Forces

- Intramolecular** forces (also known as chemical bonds) hold atoms together in a molecule:

- covalent bonds
- ionic bonds
- metallic bonds

- Intermolecular** forces (IMF) are attractive forces between molecules

- Dispersion forces
 - dipole - dipole forces
 - hydrogen bonding
 - ion - dipole forces
- often referred to as van der Waals forces

All electrostatic by nature charge

$$E = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

distance

stronger
↓

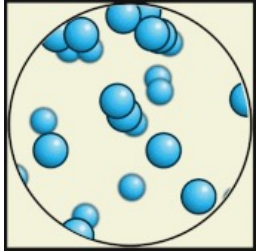
- In general, **Intramolecular** forces is stronger than **Intermolecular** forces
- Breaking **Intramolecular** forces is a chemical change. chemical rxn
- Breaking **Intermolecular** forces is a physical change:

phase change; dissolution.

(s) → (l) → (g)

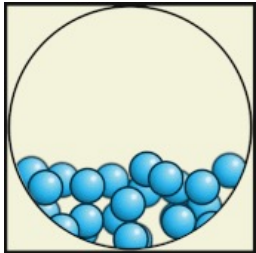
(s) → (aq)
(l) →
(g) →

Physical States of Matter



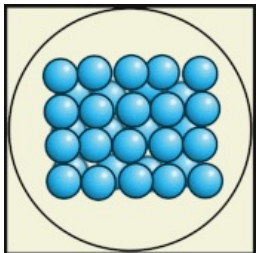
Gases

- *Widely spaced* particles-free to move around
- indefinite Volume; indefinite Shape
- low Density



Liquids

- Particles are relatively *free to move* around
- definite Volume; indefinite Shape
- high Density



Solids

- Particles packed to each other in *fixed locations*
- definite volume and definite shape
- high Density

Intermolecular
forces increase



Physical States of Matter

$$d = \frac{m}{V}$$



Gases

- *Widely spaced* particles-free to move around
- indefinite Volume; indefinite Shape
- low Density



Liquids

- Particles are relatively *free to move* around
- definite Volume; indefinite Shape
- high Density



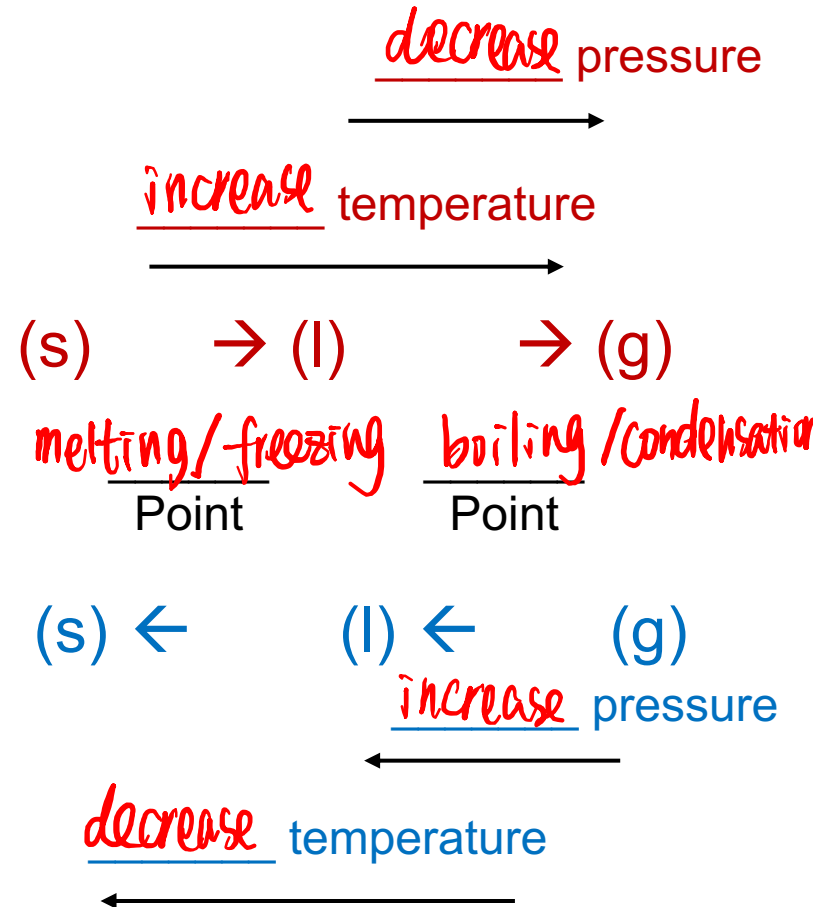
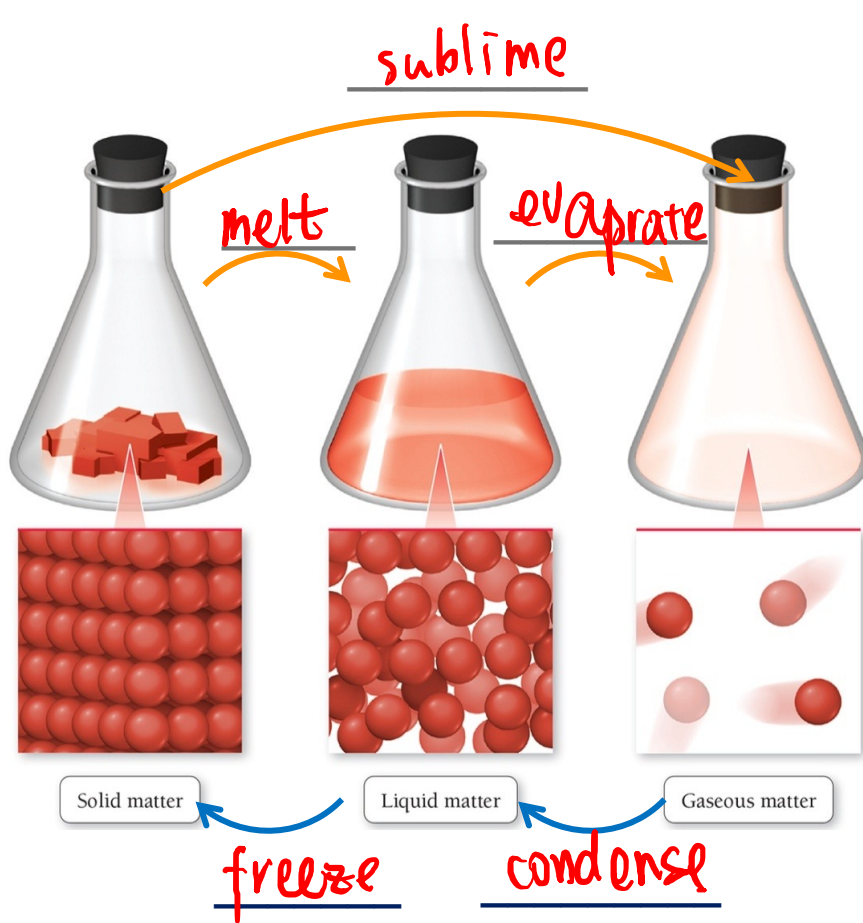
Solids

- Particles packed to each other in *fixed locations*
- indefinite volume and indefinite shape
- high Density

Intermolecular
forces _____



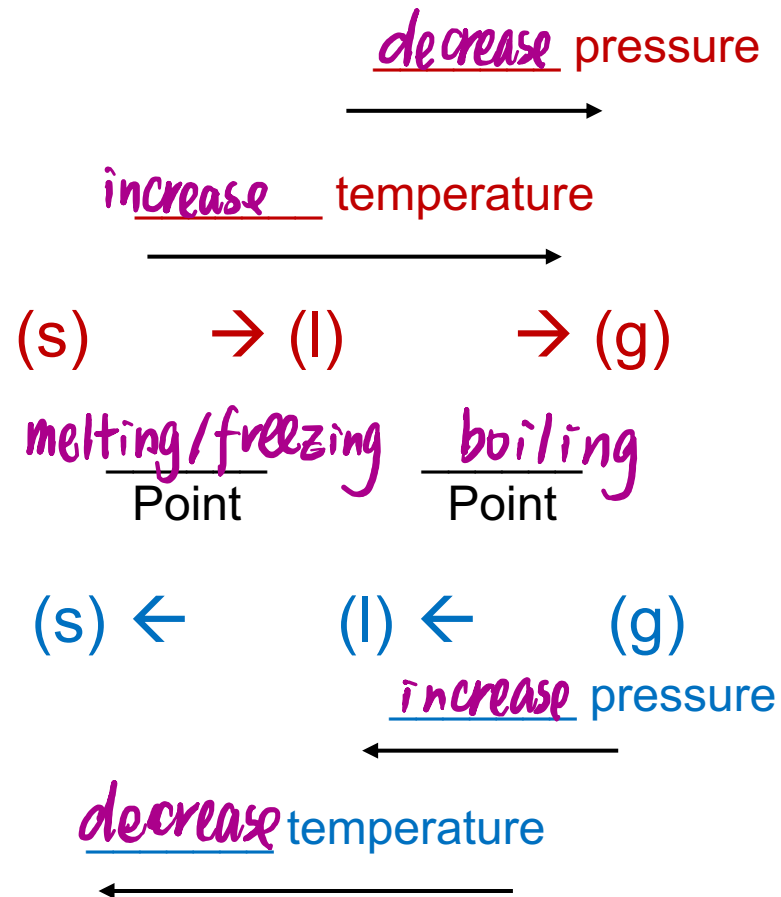
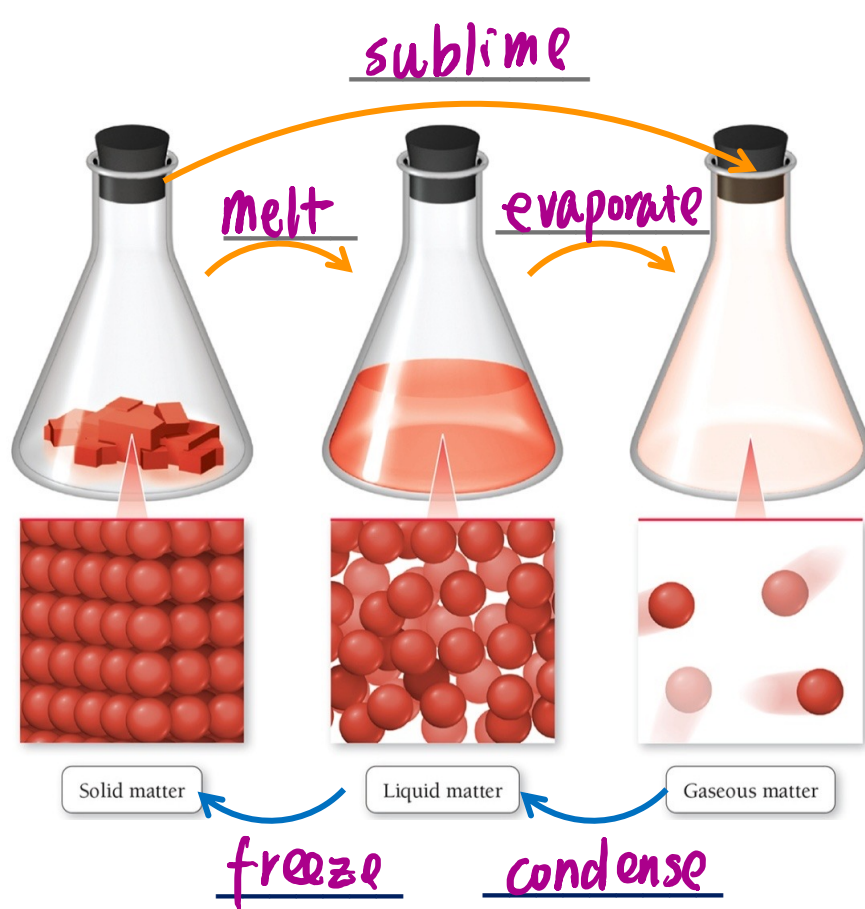
Physical States (Phase) of Matter



- Stronger intermolecular forces leads to Higher / Lower Boiling Point.



Physical States (Phase) of Matter

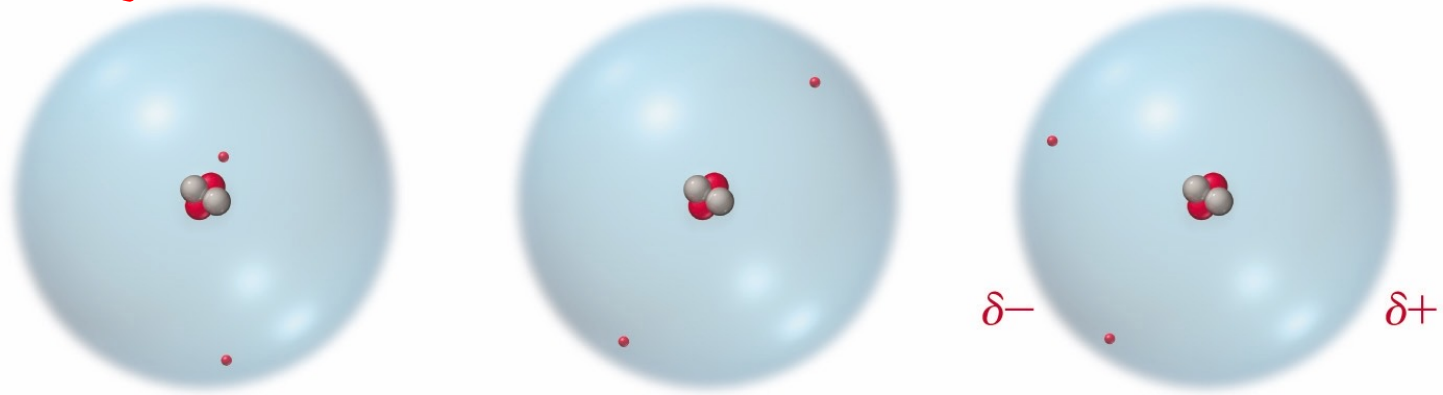


- Stronger intermolecular forces leads to Higher / Lower Boiling Point.

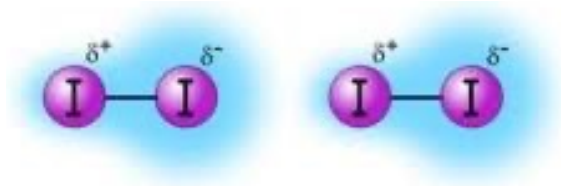


Dispersion Forces

- Dispersion forces (London forces) are attractive forces between temporary dipoles in atoms or molecules that are formed through momentary uneven distribution of electron clouds.
- Polarizability: No dipole or real charge needed.
- He:

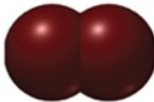
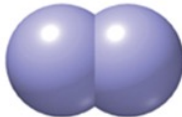
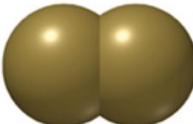


- Dispersion forces exist in **all** molecules but are most evident in nonpolar molecules.
- Strength: 0.05-20 kJ/mol
- Dispersion forces is not directional.



Dispersion Forces

- Dispersion forces (London forces) are attractive forces between temporary dipoles in atoms or molecules that are formed through momentary uneven distribution of **electron clouds**.
- larger** electron clouds, **more** polarizable; **larger** dispersion forces
- larger** molar mass will have more electrons and larger electron clouds
- Larger molar mass = stronger dispersion forces

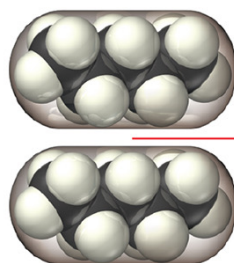
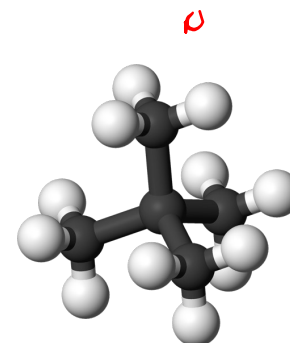
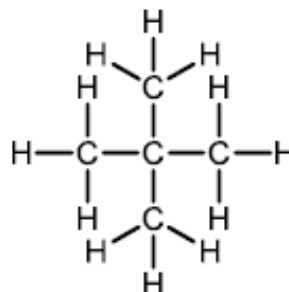
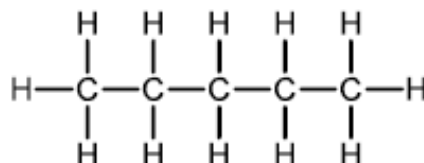
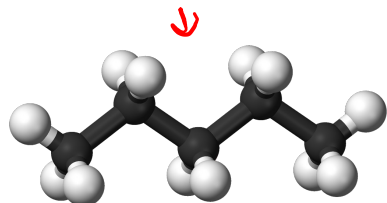
Halogen	Molecular View	Boiling Point (K)
F ₂		85
Cl ₂		239
Br ₂		332
I ₂		457
At ₂		610

Melting Points of Similar Nonpolar Compounds

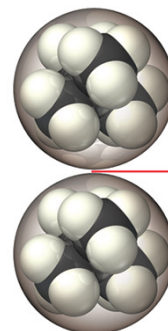
Compound	Melting Point (°C)
CH ₄	-182.5
CF ₄	-150.0
CCl ₄	-23.0
CBr ₄	90.0
CI ₄	171.0

Dispersion Forces

- Dispersion forces (London forces) are attractive forces between temporary dipoles in atoms or molecules that are formed through momentary uneven distribution of **electron clouds**.
- Larger surface area, larger dispersion forces.
- IMF (straight chains) > IMF (branched chains)



Large area for
interaction



Small area for
interaction

n-Pentane

molar mass = 72.15 g/mol

boiling point = 36.1 °C



Neopentane

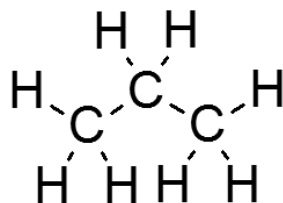
molar mass = 72.15 g/mol

boiling point = 9.5 °C

Your Turn!

- Which compound is more likely to stay in the liquid phase in room temperature?

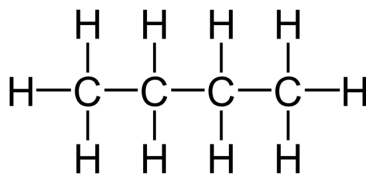
more ↓
A. C_3H_8
Propane



less Dispersion

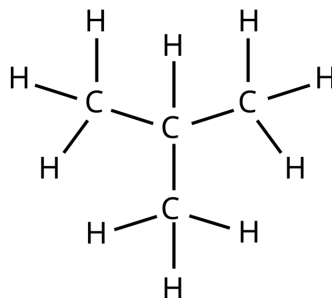


B. C_4H_{10}
n-butane



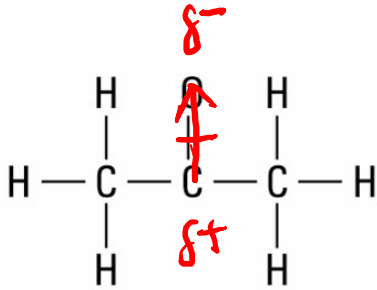
largest Dispersion

C. C_4H_{10}
Isobutane



Dipole-Dipole Forces

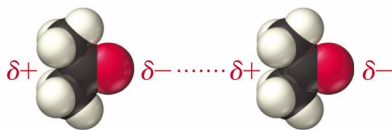
- Dipole-dipole forces are attractive forces between polar molecules (i.e., the molecules with a nonzero, permanent dipole moment).
- Strength: 3-20 kJ/mol
- Dipole-dipole forces is directional (+/- attract; +/+ or -/- repel)



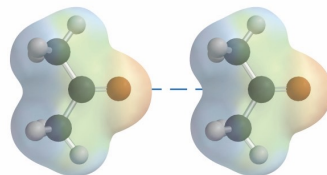
Acetone

Dipole-Dipole Interaction

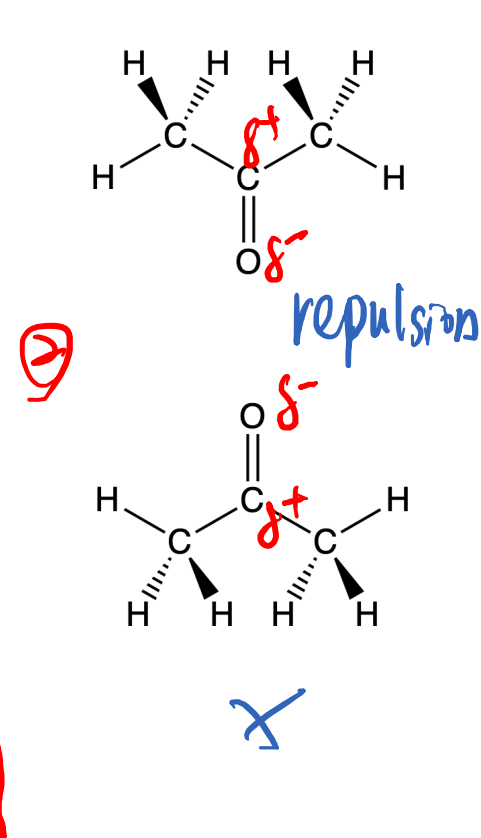
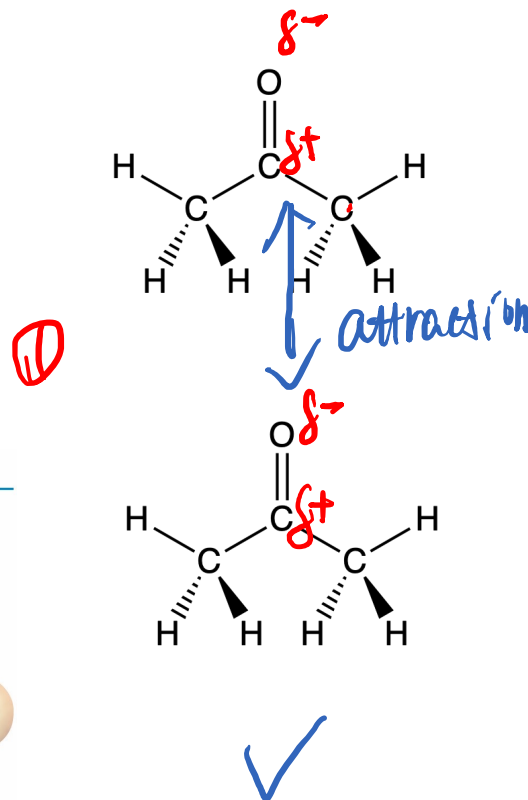
The positive end of a polar molecule is attracted to the negative end of its neighbor.



Space-filling model

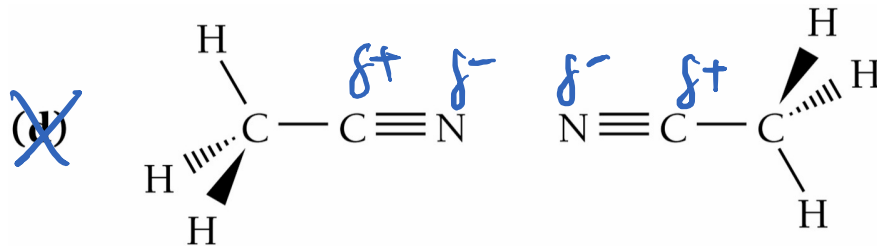
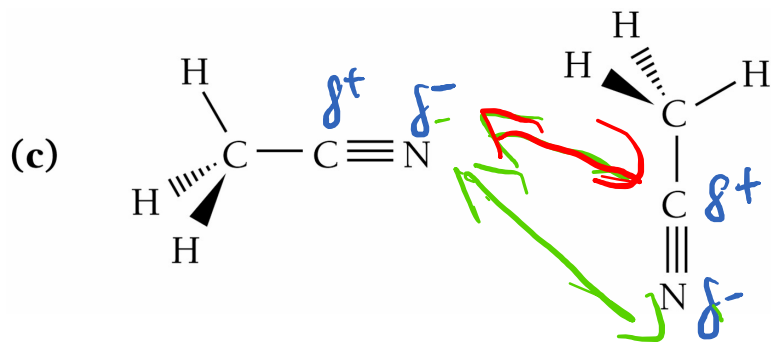
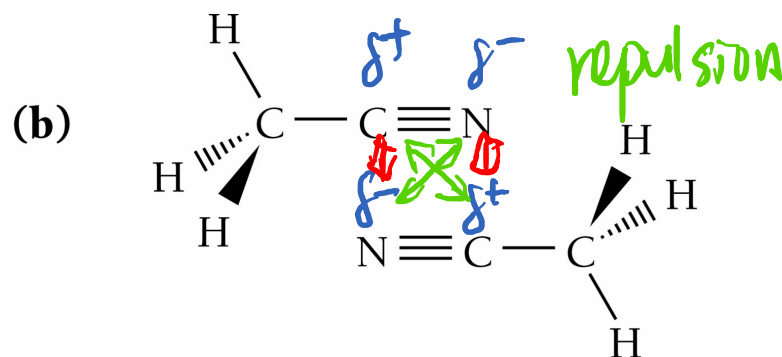
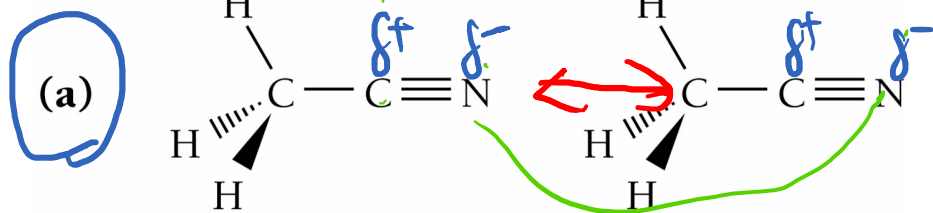
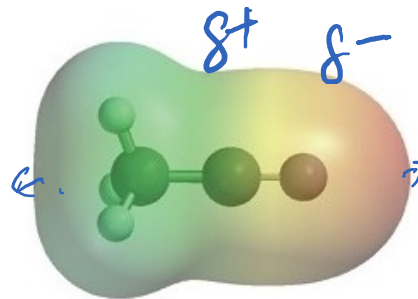


Electrostatic potential map



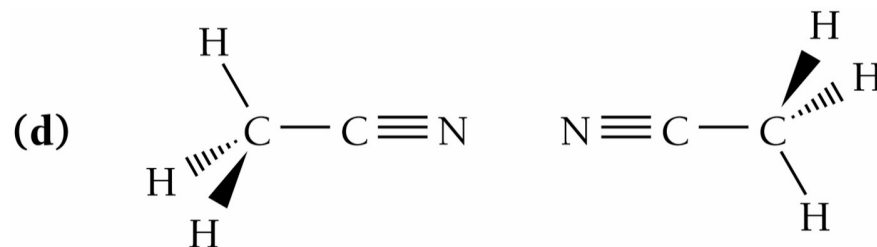
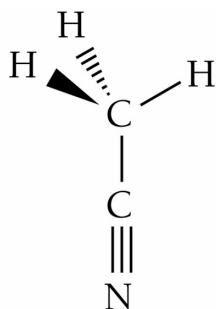
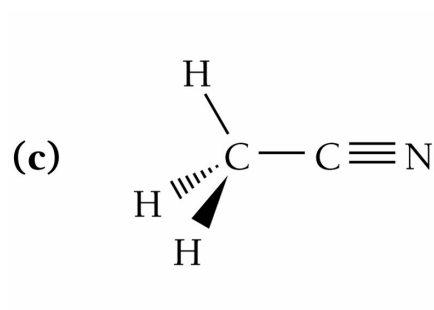
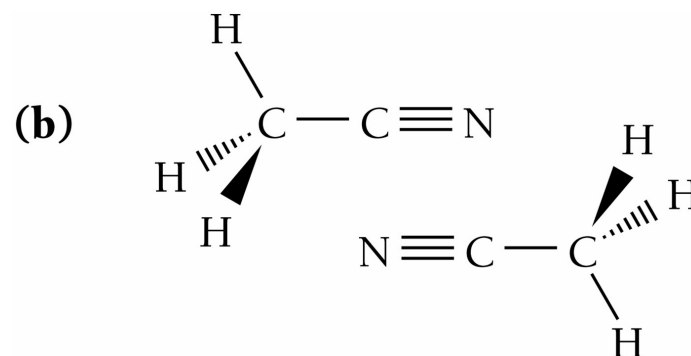
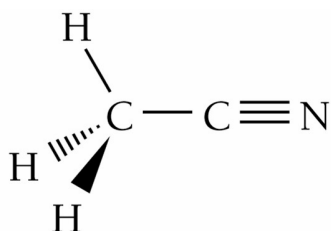
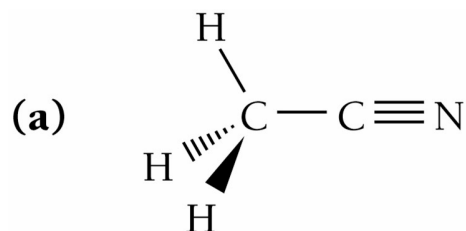
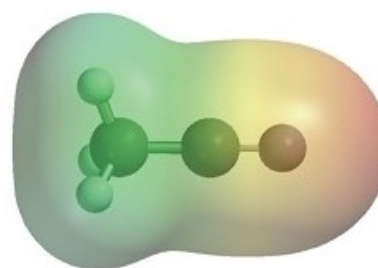
Your Turn!

An electrostatic potential map for acetonitrile (CH_3CN) which is polar, is shown below. From this map, determine the geometry for how two acetonitrile molecules would interact with each other. Which of the orientations shown here is most likely?



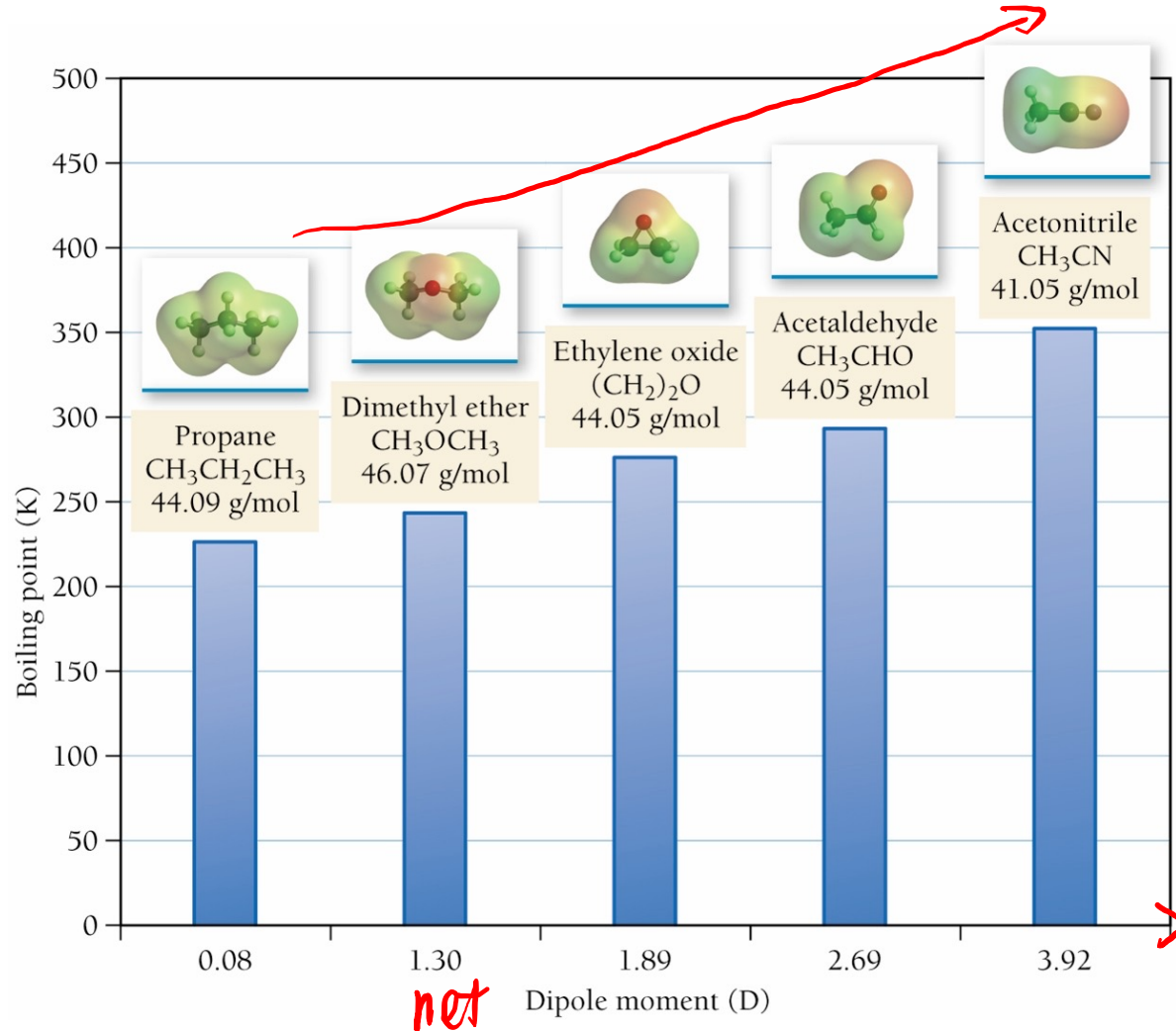
Your Turn!

An electrostatic potential map for acetonitrile (CH_3CN) which is polar, is shown below. From this map, determine the geometry for how two acetonitrile molecules would interact with each other. Which of the orientations shown here is most likely?



Dipole-Dipole Forces

- Dipole-dipole forces is stronger than dispersion forces.
- The more the dipole moment (Δ_{EN}), the greater the force.

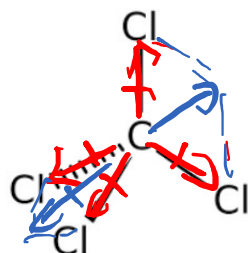


Your Turn!

Which substance experiences dipole–dipole forces?

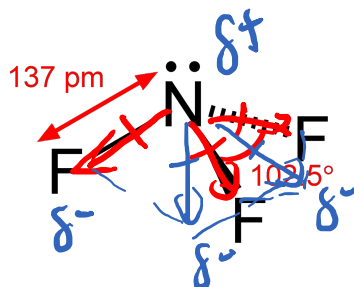


~~A.~~ CCl_4

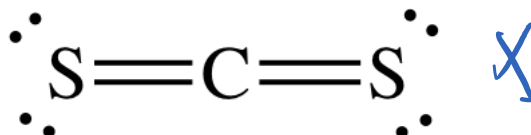


ΔEN

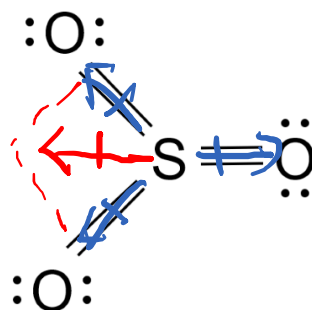
B. NF_3



C. CS_2



~~D.~~ SO_3



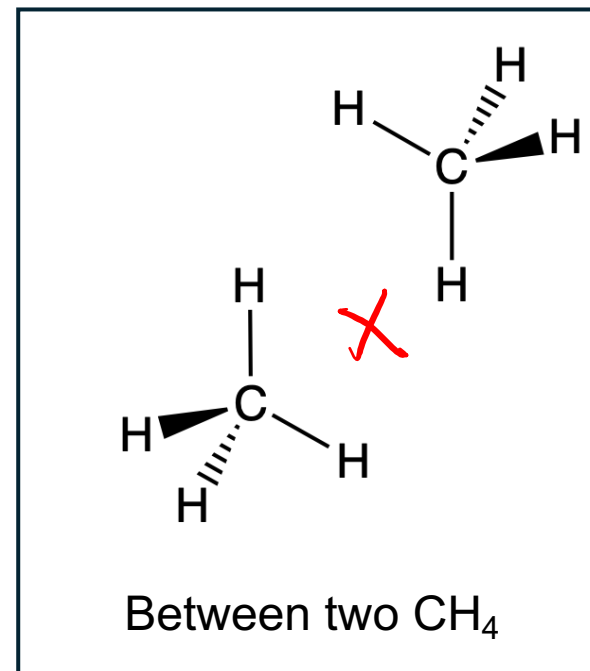
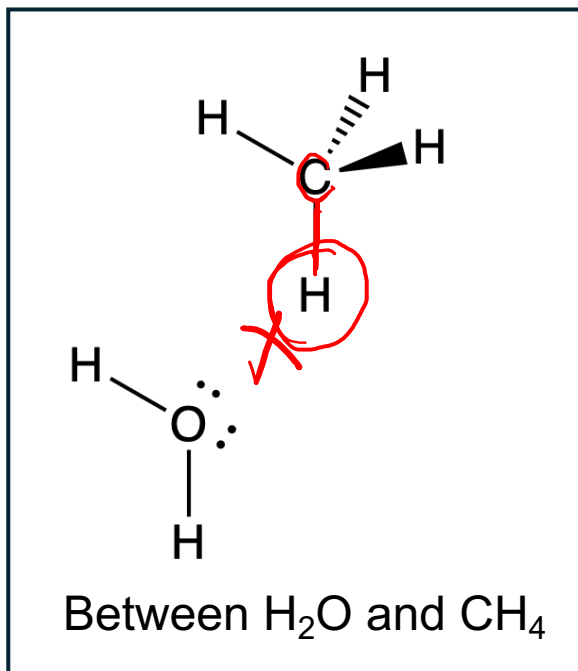
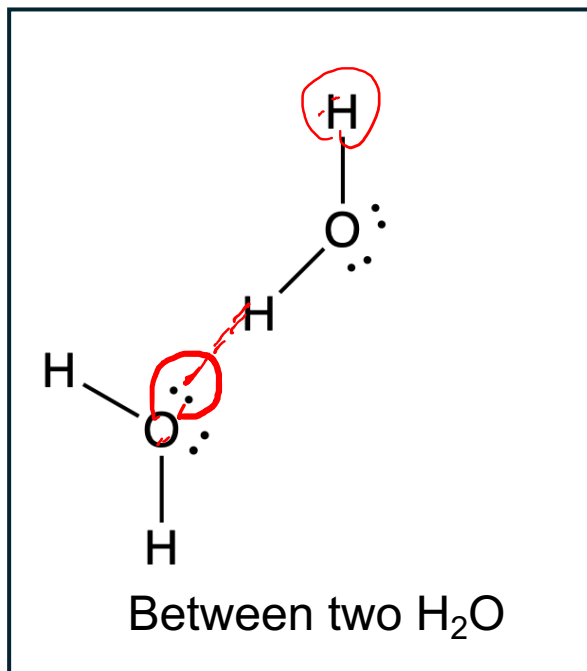
H					
2.20					
B	C	N	O	F	
2.04	2.55	3.04	3.44	3.98	
Al	Si	P	S	Cl	
1.61	1.90	2.19	2.58	3.16	
Ga	Ge	As	Se	Br	
1.81	2.01	2.18	2.55	2.96	
In	Sn	Sb	Te	I	
1.78	1.96	2.05	2.1	2.66	

Hydrogen Bonding

- A hydrogen bond is not a real chemical bond but a special type of dipole-dipole interaction.
- Hydrogen bonding is directional. ($\text{X}-\text{H} \cdots \text{X}$)

To form a H bond:

- H atom(s) present.
- The H atom is bonded to a N/O/F (highly electronegative)
- There's **another** N/O/F to attract this hydrogen. (e⁻ rich and has lone pairs)

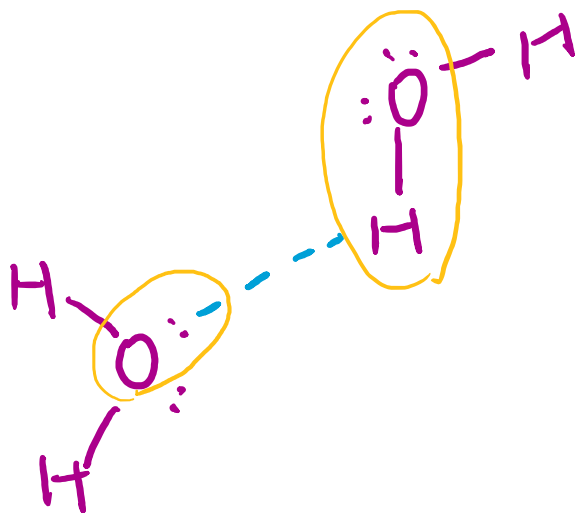


Hydrogen Bonding

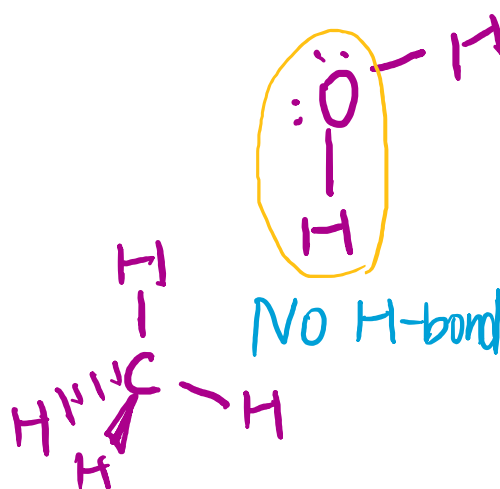
- A hydrogen bond is NOT a real chemical bond but a special type of dipole-dipole interaction.
- Hydrogen bonding is directional. (X—H · · · X)

To form a H bond:

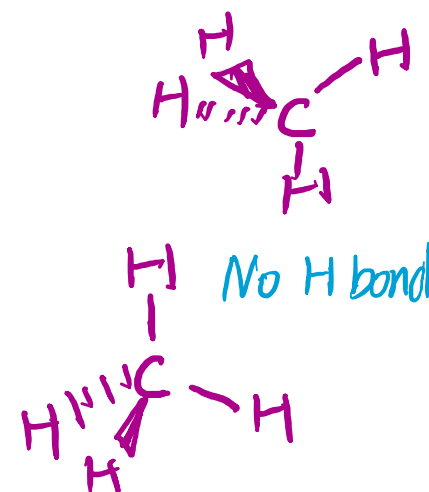
- H atom(s) present.
- The H atom is bonded to a N/F/O (highly electronegative)
- There's **another** N/F/O to attract this hydrogen. (e⁻ rich and has lone pairs)



Between two H₂O



Between H₂O and CH₄

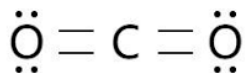
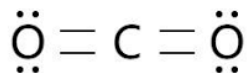


Between two CH₄

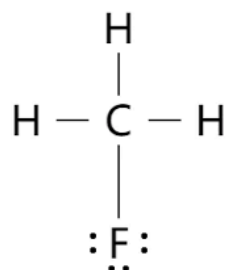
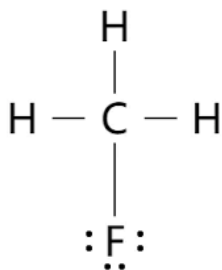
Your Turn!

In the following three pure substances, which pure substance contains hydrogen bonds?

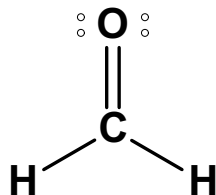
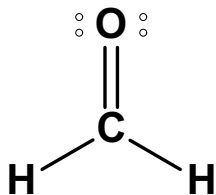
~~A.~~



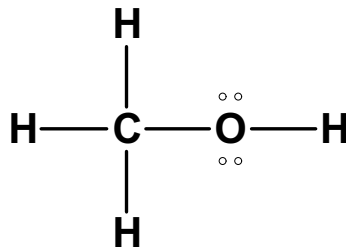
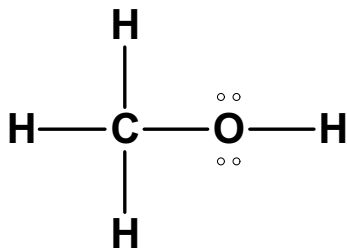
B.



C.



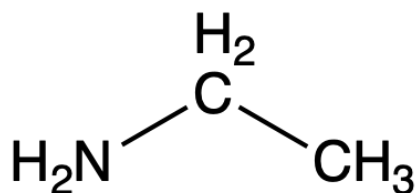
D.



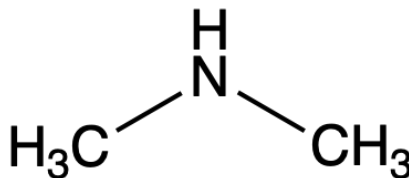
Your Turn!

How many compounds, of the ones listed below, have hydrogen bonding?

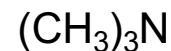
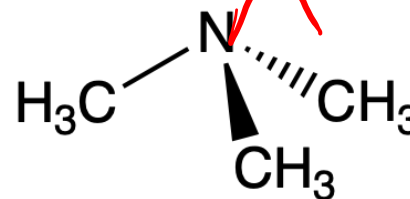
1)



2)



3) X



A) 3

B) 2

C) 1

D) 0



Announcement 3/6

Assignments that are due soon

- Pre-Class Assignment Week 6-M

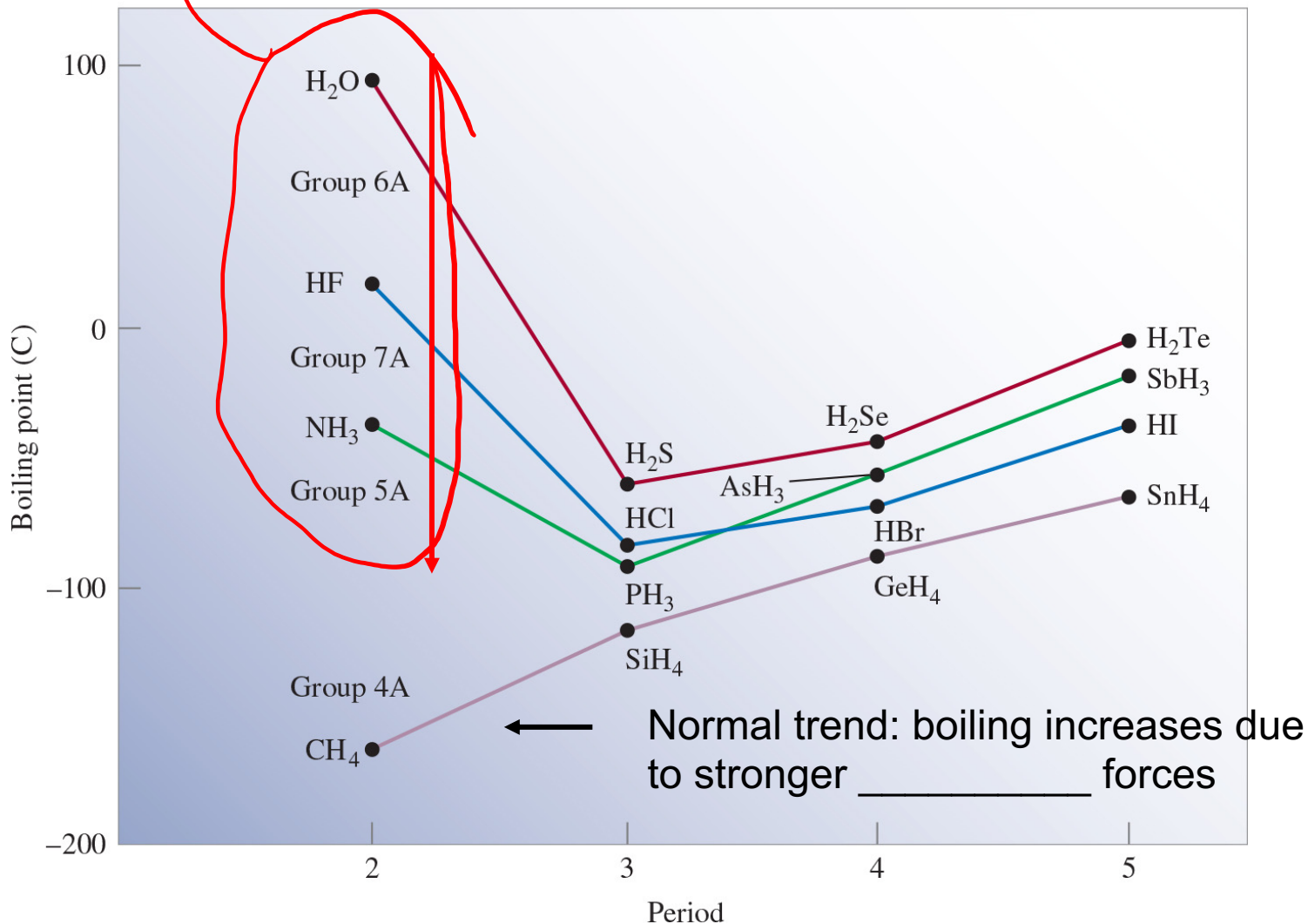
Office Hour

- Dr. Wang's: Fri 1-2 PM at KC 382



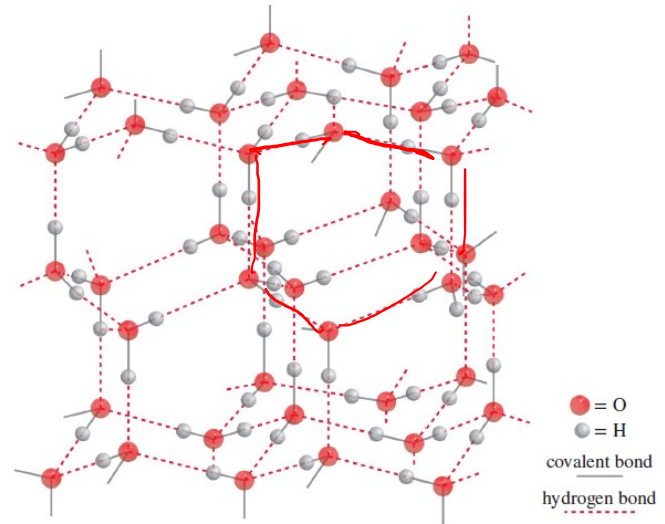
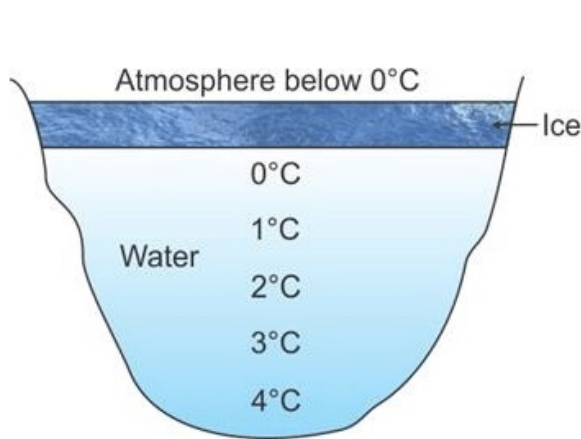
Hydrogen Bonding

- Hydrogen bonding is stronger than a typical dipole-dipole interaction.
- Strength: 10-40 kJ/mol

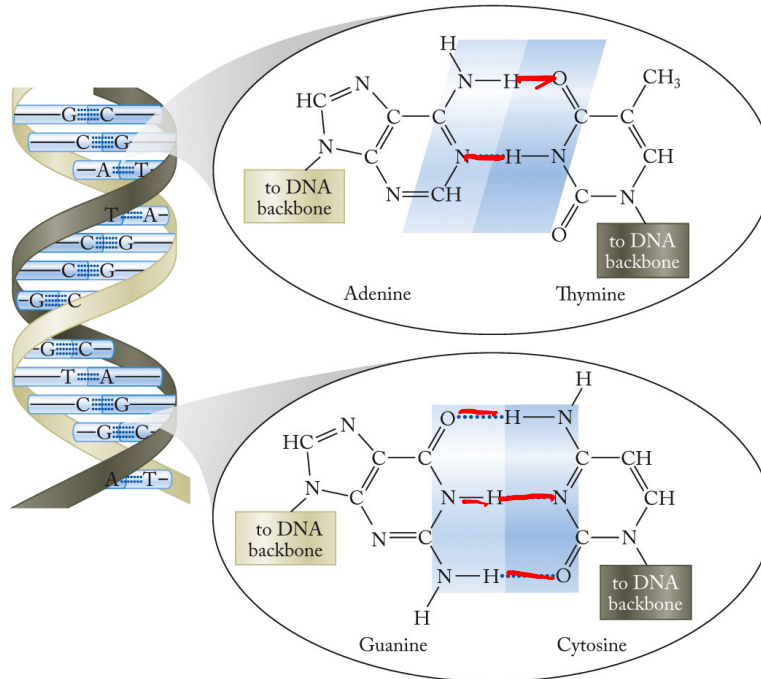


Hydrogen Bonding are everywhere!

- Density of H_2O (s) is larger than Density of H_2O (l).

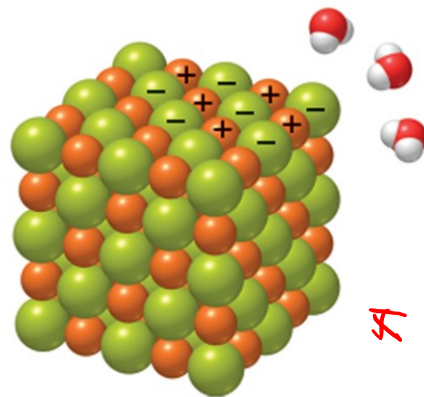
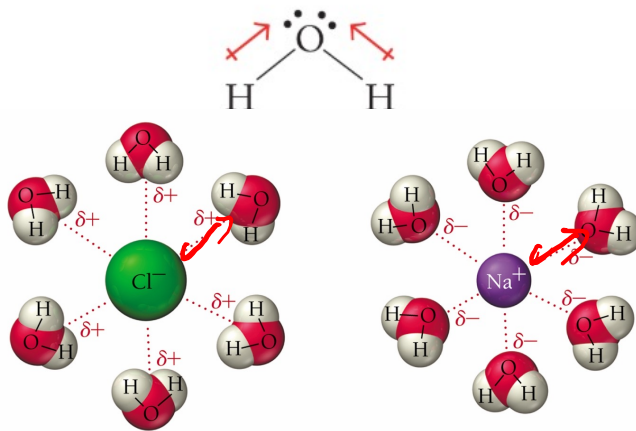


- Hydrogen bonds in DNA

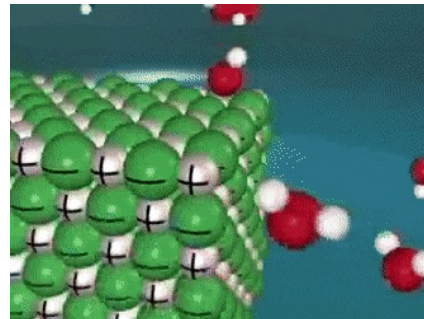


Ion-Dipole Forces

- Ion-dipole forces are attractive forces between an ion (a point charge) and a polar molecule
- Strength: 30-100 kJ/mol

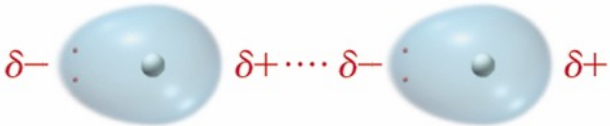
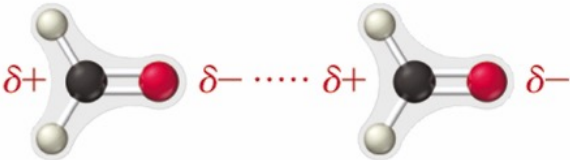
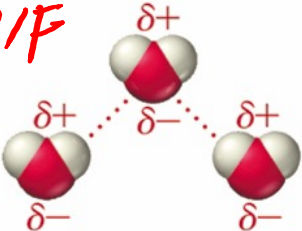
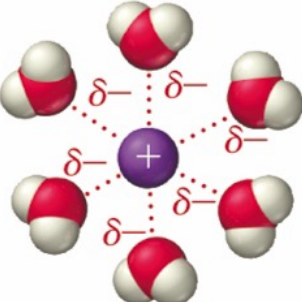


Solid NaCl



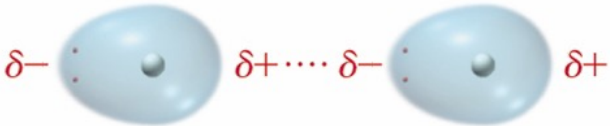
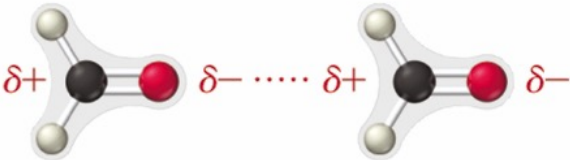
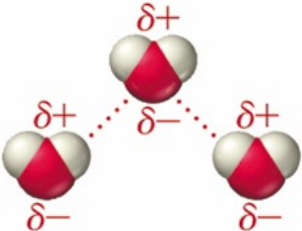
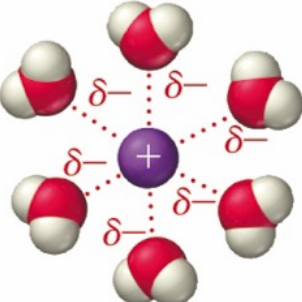
Dissolving NaCl in water

Summary of Intermolecular Forces

Type	Present In	Molecular Perspective	Strength
Dispersion	all molecules		0.05–20+ kJ/mol
Dipole–Dipole	polar molecules		3–20+ kJ/mol
H-bond	"naked" H–N/O/F F/O/N		10–40 kJ/mol
ion–dipole	ion + polar molecule solution		30–100+ kJ/mol

*The dispersion force can become very strong (as strong and even stronger than the others) for molecules of high molar mass.

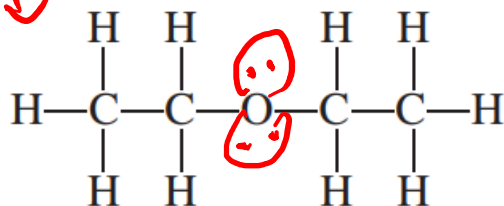
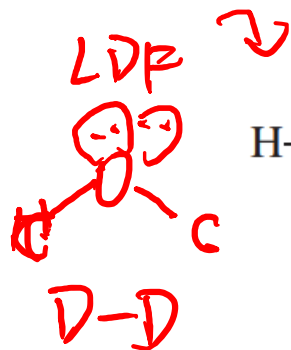
Summary of Intermolecular Forces

Type	Present In	Molecular Perspective	Strength
Dispersion*	All molecules and atoms		0.05–20+ kJ/mol
Dipole–dipole	Polar molecules		3–20+ kJ/mol
Hydrogen bonding	Molecules containing H bonded to F, O, or N		10–40 kJ/mol
Ion–dipole	Mixtures of ionic compounds and polar compounds		30–100+ kJ/mol

*The dispersion force can become very strong (as strong and even stronger than the others) for molecules of high molar mass.

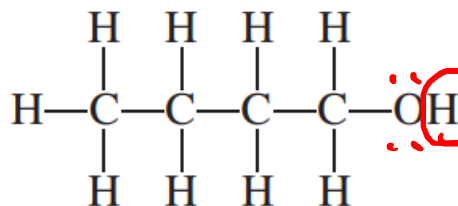
Your Turn!

Which intermolecular force leads to the difference in melting point for the two molecules below.



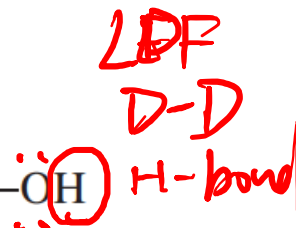
diethyl ether

mm = 74.12 g/mol
bp = 34.5°C



1-butanol

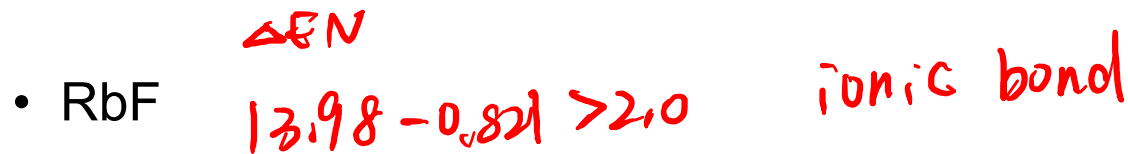
mm = 74.12 g/mol
bp = 117°C



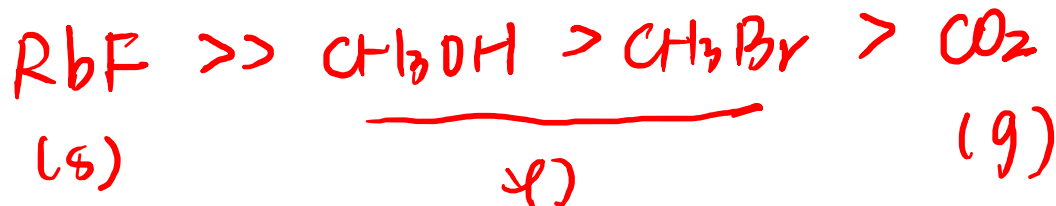
- A. Dispersion force
- B. Dipole-dipole force
- ☒ C. Hydrogen bonding
- D. Ion-dipole force

Your Turn! EN: H(2.20); Rb(.82); F(3.98); C(2.55); O(3.44); Br(2.96)

1) Identify what inter-/intra- molecular forces exist in the following substance.



2) Arrange the following substance in the order of increasing boiling point and predict which substance is most likely to be in the gas phase at room temperature. Explain your answer.



Your Turn!

Arrange the following substance in the order of increasing boiling point:

RbF, CO₂, CH₃OH, CH₃Br.



dispersion only < (dispersion + dipole-dipole) < (dispersion + dipole-dipole + H bonding) < ionic bonds

CO₂ is most likely to be in the gas phase because it only has dispersion forces between them, which is the weakest intermolecular forces